

Chemical equilibrium and routes to K

After midterm · Session 5 of 6 · 180 minutes

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The question for today

Why can composition change while the equilibrium constant stays the same?

Before class: 3-page reading · 15–20 minutes

Bring the three preparation answers. Revisit your prediction after one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion and exit question	30

The 70-minute investigation is student work with instructor support.

Start with extent and driving force

$$n_i = n_{i,0} + \nu_i \xi, \quad n_i \geq 0$$

$$\left(\frac{dG}{d\xi} \right)_{T,P} = \Delta_r G = \Delta_r G^\circ + RT \ln Q$$

Write the feasible extent interval first. A negative derivative favors increasing extent; a positive derivative favors decreasing extent.

An interior equilibrium has zero derivative. Boundary minima need one-sided checks.

The equilibrium condition gives **K**

$$\Delta_r G = 0 \Rightarrow Q = K, \quad \ln K = -\frac{\Delta_r G^\circ}{RT}$$

Obtain standard Gibbs energy from direct data, formation properties, or H–TS at the evaluation temperature.

K depends on T and the reaction/standard-state convention. Q describes the current composition and pressure.

Additional K(T) routes are in the appendix and lab.

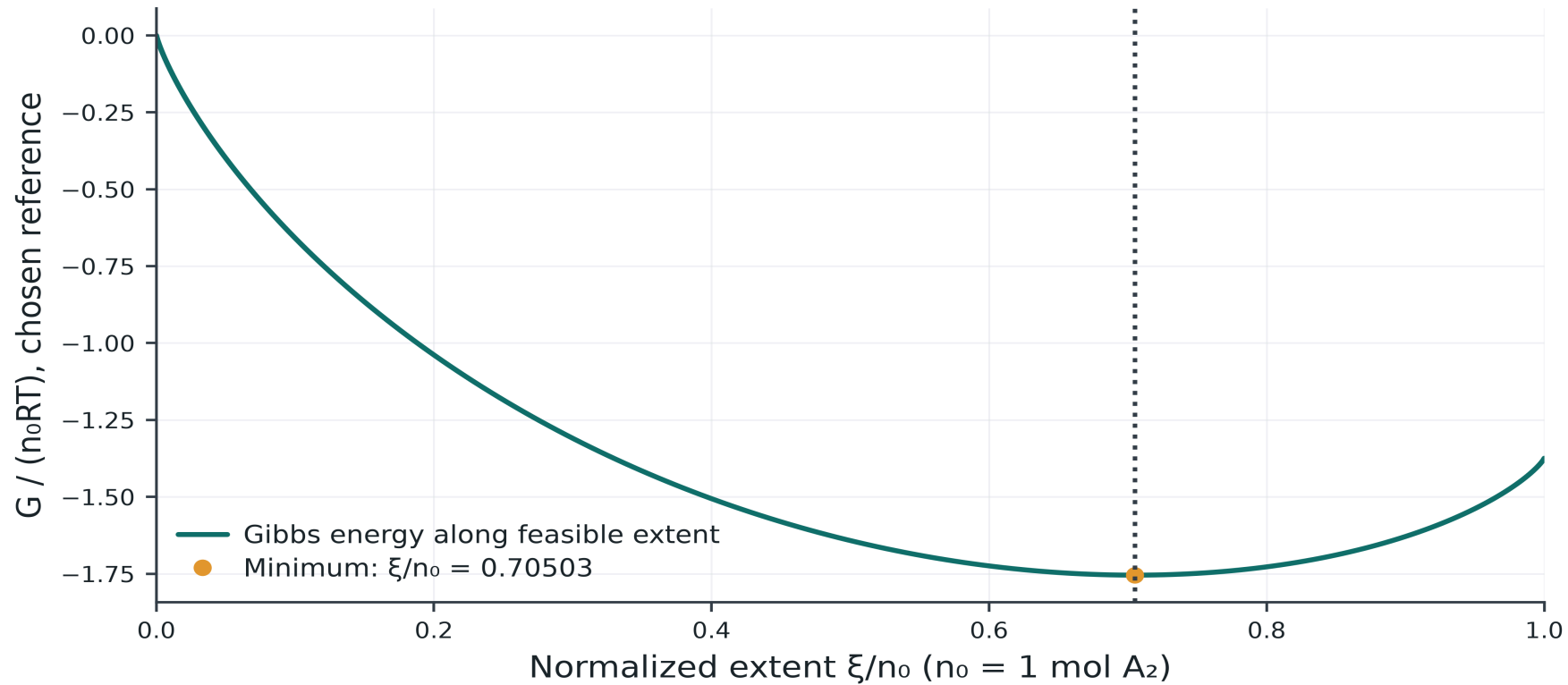
A reference-state check

Lab 12 synthetic $A_2 \rightleftharpoons 2A$ at 350 K uses $\Delta_r G^\circ = -4.000$ kJ/mol.

$$\ln K = \frac{4000}{R(350)} = 1.37454, \quad K \approx 3.95326$$

With $\Delta_r H^\circ = 50.000$ kJ/mol and $\Delta_r S^\circ = 54000/350$ J mol⁻¹ K⁻¹, the H-TS route gives the same value.

Check the minimum and the reaction quotient



Synthetic $A_2 \rightleftharpoons 2A$ at 350 K and $P=P^\circ$. Initial amount: one mole A_2 .

At $\xi \approx 0.70503$ mol, $Q \approx 3.95326 = K$. Check the atom balance.

One demonstration • 20 minutes

Open the core lab

Use the synthetic Lab 12 case. Write $n_{A_2}=1-\bar{\xi}$, $n_A=2\bar{\xi}$ and $0\leq\bar{\xi}\leq 1$ mol first. Compare the supplied Gibbs-energy and H-TS routes, then change only pressure and inspect K and extent.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 12, compare the supplied Gibbs-energy and H–TS routes at the reference temperature. Change pressure at fixed T and compare K with equilibrium extent. Check one balance.

Discussion and exit question • 30 minutes

Use 20 minutes to compare results and discuss unexpected behavior.

Why can composition change while the equilibrium constant stays the same?

Use the final 10 minutes to explain your answer individually and name one assumption that limits it.

Follow-up practice • 30–40 minutes

In Lab 12, compare the supplied Gibbs-energy and H–TS routes at the reference temperature. Change pressure at fixed T and compare K with equilibrium extent. Check one balance.

Assessment 5 and independent checkpoint

Retain one study JSON, one comparison and a short explanation. Continue from your classroom case. The hand checkpoint is included. With pre-class reading, independent work totals 45–60 minutes.

Optional study

Choose one extension: vary temperature with and without a heat-capacity correction, reverse the reaction, or scale its coefficients consistently.

[Session appendix](#) · [Module reference](#)

[Learning path](#) · [Offline package](#)

Optional work is additional. Complete the required investigation first.